



MASTER'S DEFENCE

Breeding Climate-Resilient Wheat for Nepalese Agricultural Systems Using an Integrated Analysis of Stability, Stress Tolerance and Genetic Diversity

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BACKGROUND



- One of the most important cereal crop in the world (FAOSTAT, 2024)
- Third most important cereal of Nepal (MOALD, 2023)
- Cultivated in every districts of Nepal (MOALD, 2023)
- Wheat statistics (2022/23) ~7 lakh hectares ~2 million metric tons

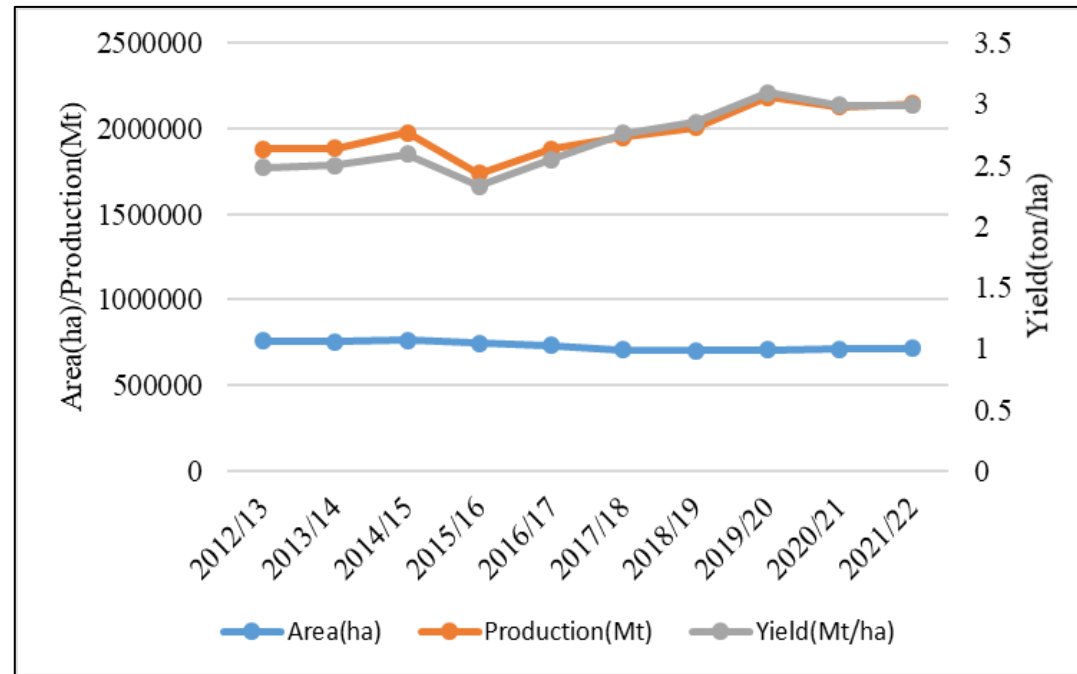


Fig. Total production, productivity and area under wheat cultivation in Nepal from 2012/13 to 2021/22 (MOALD, 2022)



STATEMENT OF THE PROBLEM



- Area under Rainfed: 51% (Pandey *et al.*, 2022)
- Area under Heat Stress: 20-27% (Tiwari *et al.*, 2019)
- Yield reduction under
 - Heat stress condition 44-47% (Poudel *et al.*, 2020)
 - Drought 50-60% (Zhao *et al.*, 2020; Poudel *et al.*, 2020)- (0.65 million metric tons)
 - Extreme drought up to 86% (Bhandari *et al.*, 2024)
- Lack of climate resilient wheat genotypes
- On Monetary term annual loss of 1.84 and 1.24 million US \$ with cumulated 3.08 million US \$ due to heat stress and drought

- Temperature rise @ 0.06°C per year (Paudel *et al.*, 2020)
- Precipitation decline @ 16.09 mm (Paudel *et al.*, 2020)

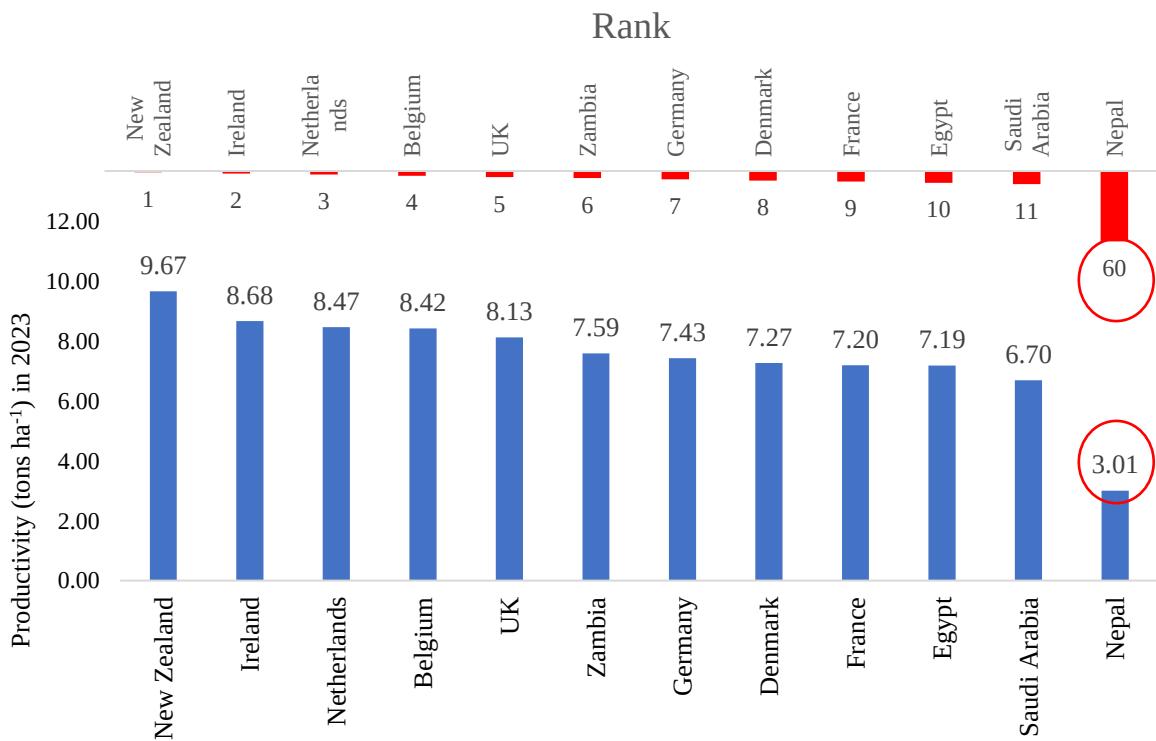
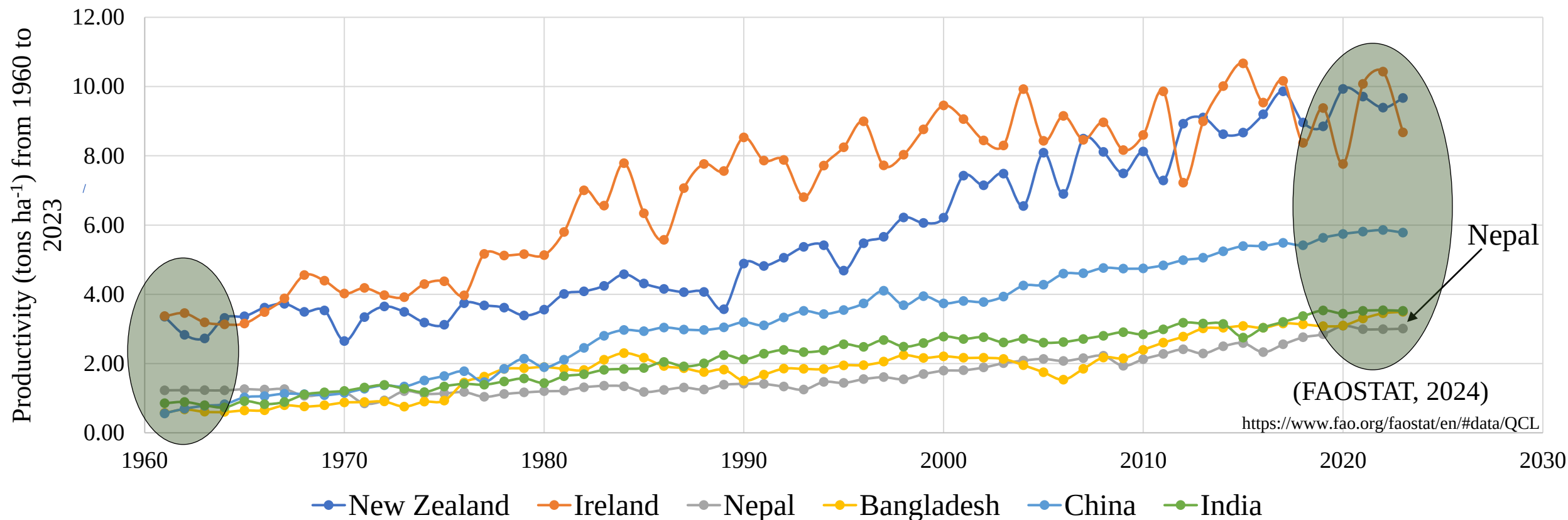


Fig. Wheat productivity of different countries (FAOSTAT, 2024)



STATEMENT OF THE PROBLEM



Projection on climate change and wheat production indices:

Temperature ↑ by 1.07 °C by 2045 AD (Djanaguiraman *et al.*, 2020, Kamrani *et al.*, 2017)

Drought and heat stress are predicted to reduce wheat yield by 44-47% in south Asia (Abhinandan *et al.*, 2017, Lesk *et al.*, 2016)

Nepal: 42 wheat varieties till 2024 (mean genetic yield potential: 4.47 t ha⁻¹); Actual yield: 3.01 t ha⁻¹ Yield gap: 1.46 t ha⁻¹ (32.66% reduction) (SQCC, 2024; FAOSTAT, 2024)



OBJECTIVES



Broad objective:

- To investigate the stability, adaptability and genetic diversity of wheat genotypes

Specific objectives:

- To identify genotypes with drought and late sown heat tolerance in the field (Phase I)
- To identify potential traits for trait based selection of high yielding wheat genotypes
- To validate Mean Stable Yield (MSY) as a potential selection indices for extreme stress conditions
- To identify genotypes with heat and drought based SSR genetic marker for climate resilient breeding (Phase-II)



HYPOTHESIS



Null Hypothesis:

- There is no significant genotypic and environmental diversity across tested wheat genotypes and environments
- There is no significant genetic diversity across tested genotypes with reference to moisture and heat-based SSR markers



RATIONALE OF THE STUDY



- **Food and nutritional security**
 - 16.67% of Nepalese suffer from some forms of malnutrition and 6% suffer from chronic micronutrient deficiency (MOF, 2022).
 - Improvement of position of Nepal in Global Hunger Index 2022 [current (81th)]
 - Sustainable Development Goal 2.0 (Eradication of all forms of hunger and malnutrition by 2030)
- **16th Five year plan**
 - Reducing households in high food insecurity from 1.3% (2022/23) to 1.0% (2028/29) [*Target on social justice*]
 - Increasing average productivity of agriculture commodities (major crops) from 3.3 t ha⁻¹ (2022/23) to 3.7 t ha⁻¹ (2028/29) (*Target on Prosperity*)



MATERIALS AND METHODS



Experimental phases (I)

On-Field Evaluation

- 21 wheat genotypes
- 4 Bhairahawa lines
- 13 Nepal lines
- 4 Commercial check varieties
- RCB
- 3 replicates, 21 genotypes
- Plot size: 3*1.5 m²
- Date of sowing: November 14th 2024 (IR, RF, D)
Date of sowing: December 26 (LD)
- Drought tolerant check: Bhrikuti
- Heat stress tolerant check: Gautam
- Bio-fortified check: Zinc Gahu-1
- Environments: 4 (IR, RF, D, LD)

S.N.	Genotypes	S.N	Genotypes
1	BL 5106	11	NL 1447
2	BL 5099	12	NL 1445
3	BL 4984	13	NL 1506
4	Bhirkuti	14	NL 1504
5	NL 1437	15	NL 1503
6	NL 1402	16	NL 1501
7	Gautam	17	Vijaya
8	BL 5116	18	NL 1512
9	NL 1492	19	NL 1509
10	NL 1488	20	NL 1508
		21	Zinc Gahu-1

Source: National Wheat Research Program (NWRP), Bhairahawa

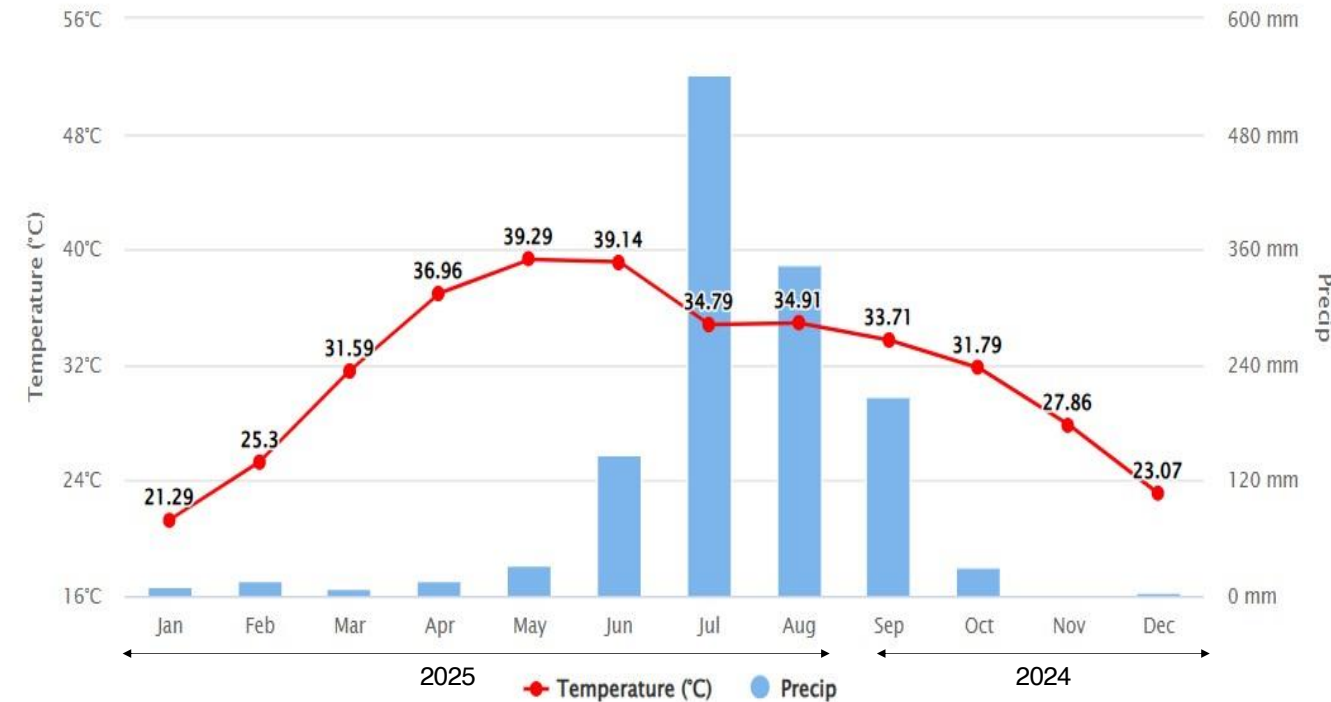


MATERIALS AND METHODS



Agro-climate and the validation of late sown rainfed (heat stress)

Late sown condition suffered from terminal heat wave of March (>24.5 °C) (Oyewole, 2016)



Validation of Rainfed and Drought Environment

	Irrigated	Rainfed	Drought	Late sown Rainfed
Pre-sowing	12.5%	12.5%	12.5%	10.8%
CRI	11.3%	10.6%	9.8%	9.8%
Booting	13.8%	9.1%	6.3%	5.6%
Heading	16.8%	13.2%	4.9%	4.8%
Soft dough	17.2%	15.1%	4%	4.1%
Harvesting	9.6%	7.2%	2.9%	1.4%

Gravimetric Method was used for the moisture estimation.



STATISTICAL ANALYSIS



S.N	Software	Analysis
1.	Ms Office (Word, Excel)	Data entry, processing, and stress tolerance indices (STI)
2	SPSS	Combined Analysis of Variance (ANOVA)
3.	GEA-R Version-4 Provided by CIMMYT Mexico	Additive Main Effect and Multiplicative Interaction (AMMI), Genotype, Genotype*Environment Interaction (GGE)
4	Past 4.13	Principal component analysis (PCA) UPGMA Based Clustering (Molecular Part)
5	R-Studio 4.6.0	Heatmap for varietal performance





STRESS TOLERANCE INDICES (STIs)

S.N.	Indices	Abbreviations	Statistical formula	References
1	Yield stability index	YSI	$\left(\frac{Y_{Si}}{Y_{pi}}\right)$	(Fischer and Maurer, 1978)
2	Stress susceptibility index	SSI	$\left[\left(1 - \frac{Y_{Si}}{Y_{pi}}\right) / \left(1 - \frac{Y_S}{Y_p}\right)\right]$	(Fischer and Maurer, 1978)
3	Mean Productivity	MP	$\left(\frac{Y_{pi} + Y_{si}}{2}\right)$	(Bouslama and Schapaugh, 1984)
4	Relative efficiency index	REI	$\left[\left(\frac{Y_{Si}}{Y_S}\right) \times \left(\frac{Y_{pi}}{Y_p}\right)\right]$	(Hossain et al., 1990)
5	Geometrical Mean productivity	GMP	$(\sqrt{Y_{pi} \times Y_{si}})$	(G. C. J. Fernandez, 1992)
6	Mean Stable Yield	MSY	$\left[\left(\frac{Y_{Si}}{Y_{pi}}\right) \times \left(\frac{Y_{pi} + Y_{si}}{2}\right)\right]$	(Bhandari et al., 2024b)
7	Stress tolerance index	STI	$\left(\frac{Y_{pi} \times Y_{si}}{Y_p^2}\right)$	(G. C. J. Fernandez, 1992)
8	Modified stress tolerance index 1	MSTI1	$\left[\left(\frac{Y_{pi}^2}{Y_p^2}\right) \times STI\right]$	(Farshadfar and Sutka, 2002)
9	Modified stress tolerance index 2	MSTI2	$\left[\left(\frac{Y_{si}^2}{Y_S^2}\right) \times STI\right]$	(Farshadfar and Sutka, 2002)
10	Harmonic mean productivity	HMP	$\frac{2 \times (Y_{pi} \times Y_{si})}{(Y_{pi} + Y_{si})}$	(Dadbakhsh et al., 2011)
11	Percentage yield loss	PYL	$\left(\frac{Y_{pi} - Y_{si}}{Y_{pi}}\right) \times 100\%$	(Sahar and Nsarellah, 2017)

Yield of genotypes under irrigated (Y_p), yield of genotypes under stress (heat, rainfed, drought) (Y_s), mean productivity (MP), geometric mean productivity (GMP), harmonic mean productivity (HMP), mean stable yield (MSY), yield stability index (YSI), relative efficiency index (REI), stress tolerance index (STI), modified stress tolerance index 1 (MSTI1), modified stress tolerance index (MSTI2), stress susceptibility index (SSI), percentage yield loss (PYL). 11



MOLECULAR MARKERS



Study area

- Lab research at Jawaharlal Nehru University, New Delhi

Genetic material

- Seeds of 21 wheat genotypes from NWRP, Bhairahawa, Nepal

Genomic DNA extraction

- Modified CTAB method (Yu *et al.*, 2017)

Materials

- **Chemicals:** Tris-HCl, NaCl, EDTA, CTAB, β -mercaptoethanol
- **Buffer:** CTAB-based extraction buffer (with β -ME added fresh)
- **Tools:** Mortar and pestle, sand, liquid nitrogen

Marker

Primer sequence

wms46

F- GCA CGT GAA TGG ATT GGA C

R- TGA CCC AAT AGT GGT GGT CA

wms130

F- AGC TCT GCT TCA CGA GGA AG

R- CTC CTC TTT ATA TCG CGT CCC

wms262

F- AAA CCC TTG CAC AAA TC GA

R- TTT AGC TAC AGG AGG CC

Xbarc197

F: CGCATGGTCAGTTTTCTTTTAATCC

R: GCGCTCTCCTTCATTATGGTTTGTTG

Xwmc233

F-GAC GTC AAG AAT CTT CGT CGG A

R-ATC TGC TGA GCA GAT CGT GGTT

Xwmc603

F-ACA AAC GGT GAC AAT GCA AGG A

R-CGC CTC TCT CGT AAG CCT CAA C

Xwmc695

F-GAG GGC ACC TCG TAA GTT GG

R-GGC AGG AGC CCC TAC AAG AT



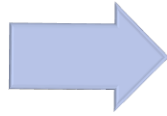
LAB PROTOCOL



PCR protocol

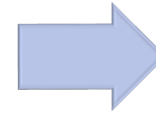
PCR setup includes:

- Template DNA: 5 μ L
- Taq polymerase buffer: 10 μ L
- Taq polymerase: 1 μ L
- dNTPs: 2 μ L
- Primers: 4 μ L
- Water: 28 μ L



Thermal cycling

- **Initial denaturation:**
94°C - 5 min
- **35 Cycles:**
94°C (30 s) $\rightarrow$ 54-57°C (30 s)
 $\rightarrow$ 72°C (1 min)
- **Final extension:**
72°C - 10 min (Hold: 4°C)



Gel electrophoresis

- 1 % agarose + 1X TBE buffer + 0.5 μ g/mL EtBr
- Sample 8 μ L, loading dye 2 μ L
- 100 Volts, 60 min
- UV transilluminator



RESULTS AND DISCUSSION

Combined ANOVA

Environment	DTH	DTM	Ph	SL	NSPMS	SPS	GPS	TSW	TGW	GY
Irrigated	84.49 ± 3.26a	119.62 ± 2.80a	88.68 ± 7.07a	8.96 ± 0.94a	255.46 ± 44.57a	14.62 ± 2.20a	35.89 ± 7.87a	21.25 ± 4.20a	47.30 ± 4.79a	3307.24 ± 844.53a
Rainfed	80.84 ± 3.23c	113.83 ± 3.81c	63.35 ± 6.54b	6.84 ± 1.06b	177.22 ± 27.82b	9.37 ± 1.90b	19.63 ± 5.37b	11.10 ± 3.49b	40.43 ± 5.07b	1349.29 ± 935.18b
Drought	79.32 ± 5.79b	112.57 ± 4.42b	61.15 ± 6.89b	5.94 ± 0.86c	157.13 ± 35.48c	7.94 ± 1.46c	14.02 ± 3.71c	7.57 ± 2.66c	38.27 ± 7.52c	497.90 ± 208.11c
Late Rainfed	69.94 ± 3.27d	94.37 ± 2.18d	54.10 ± 8.15c	6.87 ± 1.42b	159.29 ± 37.34c	9.42 ± 2.05b	17.63 ± 5.46b	8.53 ± 3.11c	31.72 ± 5.17d	591.23 ± 217.41c
Mean	78.65 ± 6.71	110.10 ± 10.07	66.82 ± 14.92	7.15 ± 1.55	187.27 ± 54.35	10.34 ± 3.18	21.79 ± 10.19	12.11 ± 6.41	39.43 ± 7.98	1436.41 ± 1302.05
CV	8.536	9.148	22.334	21.655	29.023	30.805	46.733	52.935	20.234	90.646
SEM	0.423	0.634	0.940	0.098	3.424	0.201	0.642	0.404	0.503	82.022
F-Value	***	***	***	***	***	***	***	***	***	***

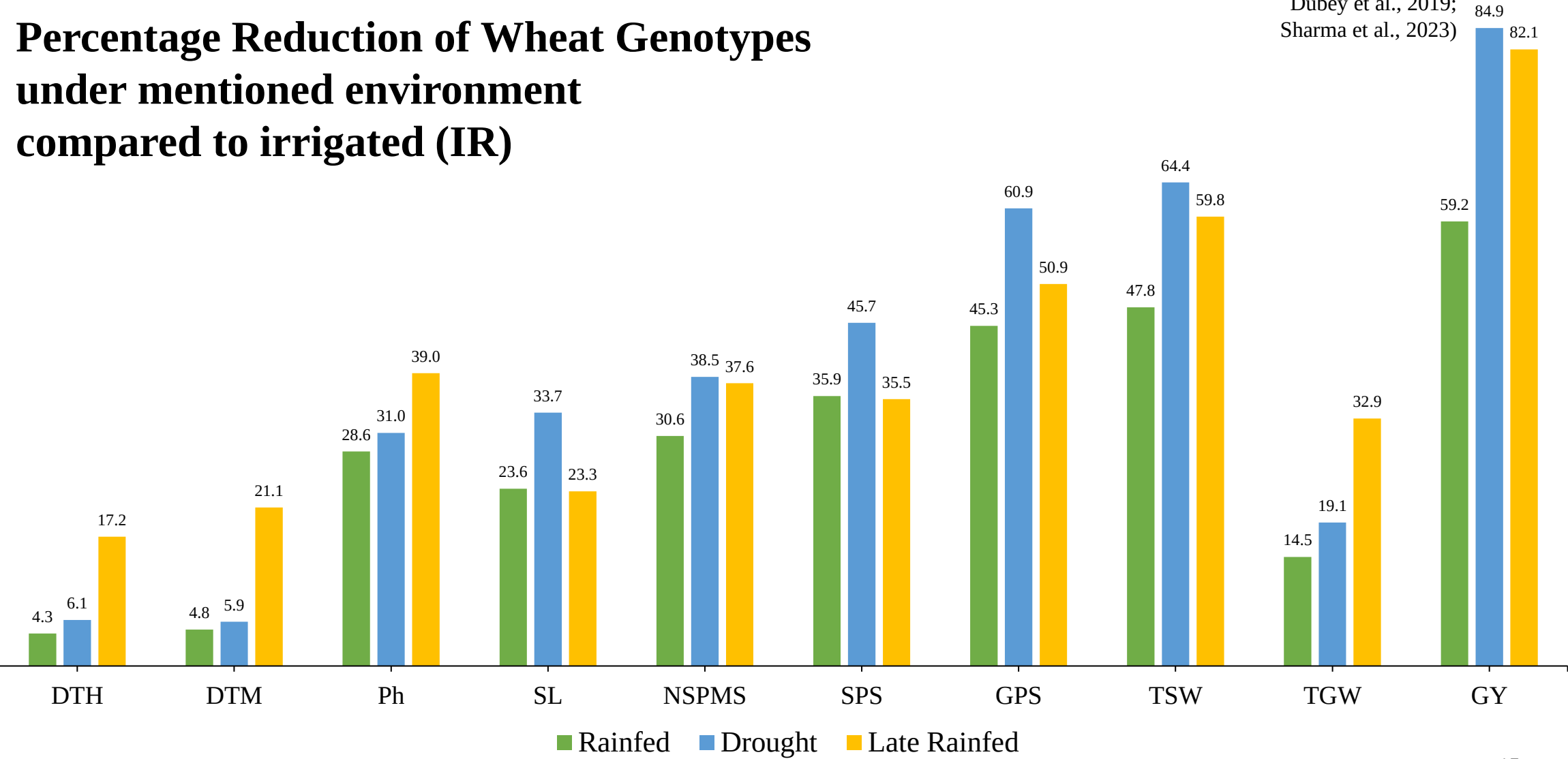


MATERIALS AND METHODS



Percentage Reduction of Wheat Genotypes under mentioned environment compared to irrigated (IR)

(Bhandari et al., 2024d, 2023;
Dubey et al., 2019;
Sharma et al., 2023)

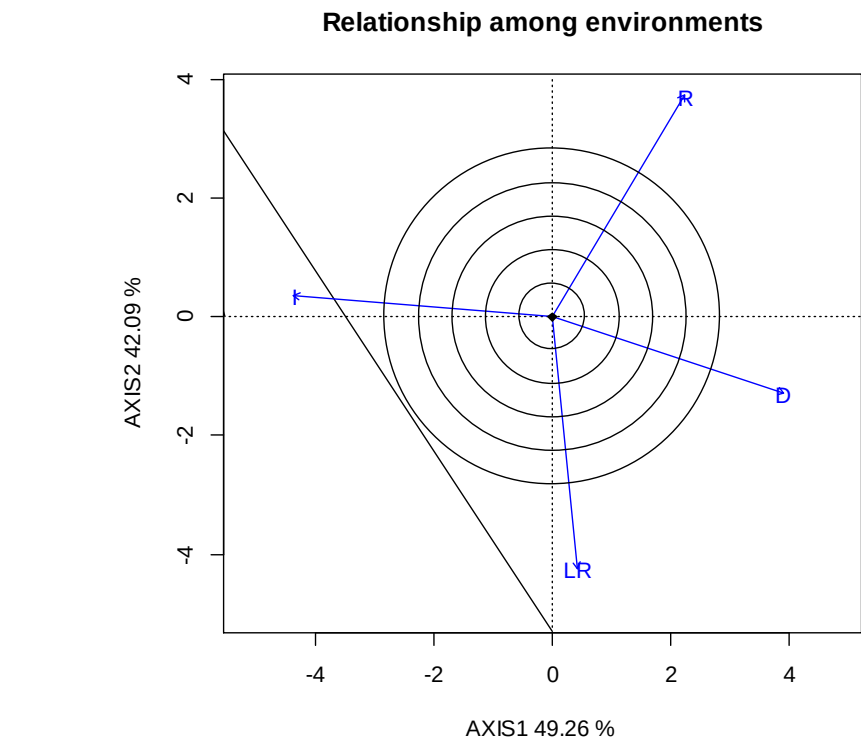




RESULTS AND DISCUSSION



	Percentage variation explained					
	ENV	GEN	G x E	PC1	PC2	PC3
DTH	70.5***	22.4***	7.1***	62.83***	26.36*	10.81
DTM	91.89***	5.23***	2.88***	50.74***	27.1*	22.16
PH	88.75***	5.97***	5.28	74.17**	18.62	7.2
SL	72.61***	11.29**	16.1	74.31**	17.38	8.3
NSPMS	74.36***	11.62***	14.02	58.74*	25.94	15.32
SPS	79.94***	7.82	12.24**	60.12	29.21	10.67
GPS	84.74***	4.57	10.7	58.06*	32.7	9.24
TSW	86.07***	4.64**	9.29	50.14*	39.74	10.12
TGW	70***	16.89***	13.11	58.95	30.95	10.1
GY	96.58***	0.94	2.49	56.48	37.95	5.57



Higher than

- Indian wheat (60-80%) (Gupta et al., 2023; Singh et al., 2019a)
- Ethiopian trials on wheat and barley (50-60%) (Derbew et al., 2024; Mulugeta et al., 2022)
- Bambara groundnut (*Vigna subterranea* L. Verdc.) (Khan et al., 2021a).



RESULTS AND DISCUSSION

Irrigated	DTH	DTM	Ph	SL	NSPMS	SPS	GPS	TSW	TGW	GY
BL 5106 (1)	88.00	120.00	92.52	9.50	244.67	17.67	44.00	23.78	44.89	3620.83
BL 5099 (2)	87.00	120.00	96.09	8.93	284.33	15.33	36.33	18.14	44.23	3551.83
BL 4984 (3)	88.00	120.00	90.40	9.70	319.00	15.33	39.00	20.71	44.14	3769.73
Bhrikuti (4)	81.00	118.67	89.89	9.73	280.33	16.00	42.00	22.69	42.34	3638.87
NL 1437 (5)	83.67	120.00	88.46	9.30	253.00	14.00	42.00	24.02	47.56	3322.97
NL 1402 (6)	84.33	119.33	86.82	9.00	281.67	14.33	35.67	19.46	41.80	3561.53
Gautam (7)	81.67	114.00	89.50	9.80	232.00	13.33	33.00	18.62	42.81	2856.00
BL 5116 (8)	85.33	120.00	83.40	8.13	301.00	12.00	30.67	18.64	46.39	2884.57
NL 1492 (9)	88.00	120.00	95.23	9.70	255.67	15.67	35.00	21.48	49.13	3410.40
NL 1488 (10)	81.00	120.00	95.83	8.73	233.33	15.33	32.33	23.07	51.07	3179.10
NL 1447 (11)	81.00	118.67	85.57	8.97	216.00	13.67	33.00	20.05	46.72	3642.37
NL 1445 (12)	88.00	125.00	91.20	8.27	239.33	14.67	40.00	25.10	49.01	3333.33
NL 1506 (13)	83.67	119.33	85.93	9.30	203.33	16.00	42.00	23.73	52.34	3503.33
NL 1504 (14)	86.67	120.00	93.57	8.87	252.00	15.00	34.33	21.87	50.75	3230.53
NL 1503 (15)	84.67	119.33	83.45	8.07	239.33	13.67	32.33	18.36	45.80	2995.67
NL 1501 (16)	80.33	116.67	86.20	8.60	264.67	13.00	33.00	19.88	48.96	3462.07
Vijaya (17)	80.33	117.33	90.84	8.40	300.67	11.33	26.00	16.01	51.07	2550.23
NL 1512 (18)	83.00	118.00	85.00	8.00	237.67	13.67	34.67	22.12	46.75	3133.57
NL 1509 (19)	88.67	123.33	82.37	9.27	240.67	15.00	34.00	22.69	50.87	3473.97
NL 1508 (20)	84.67	122.67	82.53	8.67	263.67	16.67	40.00	23.01	48.28	3287.50
Zinc Gahu 1 (21)	85.33	119.67	87.46	9.17	222.33	15.33	34.33	22.85	48.41	3043.60
Mean	84.49	119.62	88.68	8.96	255.46	14.62	35.89	21.25	47.30	3307.24
STD	2.85	2.27	4.35	0.57	30.04	1.51	4.51	2.40	3.11	311.09
CV	3.37	1.90	4.91	6.38	11.76	10.36	12.58	11.30	6.58	19.41
F-Value	***	***	ns	ns	ns	ns	ns	ns	ns	ns



RESULTS AND DISCUSSION



Rainfed	DTH	DTM	Ph	SL	NSPMS	SPS	GPS	TSW	TGW	GY
BL 5106 (1)	84.00	115.33	70.48	7.66	155.00	12.33	27.00	15.03	40.30	1598.63
BL 5099 (2)	81.67	114.67	69.03	7.10	177.33	9.33	20.00	10.64	39.35	1273.23
BL 4984 (3)	83.33	116.00	64.51	7.31	199.67	9.00	20.33	10.25	37.13	1342.37
Bhrikuti (4)	79.00	111.33	64.82	7.11	179.00	8.67	23.00	10.80	37.99	1702.40
NL 1437 (5)	81.67	116.67	61.73	7.18	182.00	8.67	18.67	11.06	42.60	1255.87
NL 1402 (6)	80.33	114.00	61.13	6.53	207.67	9.00	16.33	8.79	41.93	1355.40
Gautam (7)	79.67	111.33	73.45	8.90	154.33	11.67	26.67	17.85	43.78	1388.40
BL 5116 (8)	83.67	116.67	61.62	6.61	197.67	9.67	19.67	10.68	41.87	1602.00
NL 1492 (9)	81.00	111.33	67.79	7.60	184.33	10.00	21.00	11.27	39.16	1697.50
NL 1488 (10)	77.67	115.33	65.68	6.81	170.00	8.33	17.33	10.51	46.67	1600.70
NL 1447 (11)	79.00	113.33	63.13	6.85	156.67	9.00	19.33	10.90	40.08	1338.77
NL 1445 (12)	85.33	119.00	62.05	5.77	159.67	7.33	12.00	6.93	37.59	778.67
NL 1506 (13)	81.67	116.67	59.28	6.68	147.67	12.00	22.67	13.49	41.99	1167.57
NL 1504 (14)	86.00	118.00	59.97	6.58	192.00	9.00	21.67	12.05	40.51	1256.93
NL 1503 (15)	82.33	114.67	52.61	5.63	181.67	7.67	17.33	7.68	29.65	963.07
NL 1501 (16)	75.67	107.33	60.73	6.73	196.67	8.33	16.67	9.02	38.09	1145.13
Vijaya (17)	73.67	106.00	71.15	6.61	154.33	8.00	14.33	9.48	51.91	1127.67
NL 1512 (18)	78.33	110.00	63.65	6.48	162.00	9.00	23.67	12.91	38.42	1185.17
NL 1509 (19)	81.00	114.67	57.32	6.10	208.00	10.33	15.00	8.46	38.85	1265.30
NL 1508 (20)	81.67	115.33	56.77	6.14	191.33	9.00	16.67	10.38	42.04	1690.93
Zinc Gahu 1 (21)	81.00	112.67	63.38	7.33	164.67	10.33	23.00	14.91	39.16	1599.43
Mean	80.84	113.83	63.35	6.84	177.22	9.37	19.63	11.10	40.43	1349.29
STD	2.99	3.32	5.08	0.72	19.07	1.35	3.91	2.61	4.21	252.17
CV	3.70	2.92	8.02	10.50	10.76	14.37	19.94	23.48	10.40	18.69
F-Value	***	***	**	ns	ns	*	*	**	***	ns



RESULTS AND DISCUSSION



Drought	DTH	DTM	Ph	SL	NSPMS	SPS	GPS	TSW	TGW	GY
BL 5106 (1)	84.00	116.67	68.09	5.93	150.33	9.67	16.67	7.86	35.90	622.99
BL 5099 (2)	83.00	115.33	66.13	5.83	174.00	8.67	14.67	8.06	37.39	462.80
BL 4984 (3)	84.33	116.67	58.37	5.70	153.67	7.33	12.33	6.34	31.25	336.50
Bhrikuti (4)	71.67	107.33	59.57	6.20	176.33	8.00	14.00	6.75	35.28	427.63
NL 1437 (5)	82.00	117.00	60.53	6.33	147.67	8.00	14.33	8.39	37.00	452.43
NL 1402 (6)	78.00	110.00	58.80	5.43	166.00	7.33	13.00	6.03	33.13	429.80
Gautam (7)	75.67	110.67	65.43	6.47	179.33	7.00	13.00	7.37	40.10	478.90
BL 5116 (8)	84.67	118.00	55.97	5.30	187.00	7.00	12.67	7.14	39.31	608.40
NL 1492 (9)	79.67	111.33	62.27	6.13	172.33	8.33	14.67	8.05	34.88	549.70
NL 1488 (10)	74.33	110.67	66.43	5.27	170.33	7.33	13.67	8.47	47.36	447.97
NL 1447 (11)	72.33	108.00	59.10	5.80	151.00	7.00	12.00	6.96	42.76	531.83
NL 1445 (12)	83.00	115.33	63.77	6.03	102.33	8.00	15.67	9.21	42.72	507.07
NL 1506 (13)	75.00	109.33	64.23	6.93	153.00	9.67	17.33	9.35	40.19	531.00
NL 1504 (14)	82.33	111.33	63.30	7.10	135.33	9.33	16.67	11.00	47.52	571.93
NL 1503 (15)	83.00	115.67	56.47	5.60	155.33	8.33	15.67	7.13	35.69	432.63
NL 1501 (16)	73.00	108.00	58.50	6.17	127.00	7.67	12.33	7.23	36.97	364.27
Vijaya (17)	70.33	106.00	66.37	5.53	122.00	6.67	11.00	6.57	40.10	458.23
NL 1512 (18)	80.67	113.00	59.60	5.37	137.33	7.67	13.67	5.96	37.28	458.00
NL 1509 (19)	82.00	112.67	53.90	5.40	183.33	7.33	11.33	5.85	33.82	466.57
NL 1508 (20)	86.67	117.00	54.63	5.33	193.00	7.00	10.67	5.55	34.64	645.20
Zinc Gahu 1 (21)	80.00	114.00	62.67	6.80	163.00	9.33	19.00	9.63	40.35	672.00
Mean	79.32	112.57	61.15	5.94	157.13	7.94	14.02	7.57	38.27	497.90
STD	4.91	3.67	4.15	0.55	23.18	0.93	2.21	1.41	4.30	89.00
CV	6.19	3.26	6.79	9.34	14.75	11.78	15.77	18.68	11.24	17.87
F-Value	***	***	ns	ns	ns	ns	ns	ns	ns	ns



RESULTS AND DISCUSSION



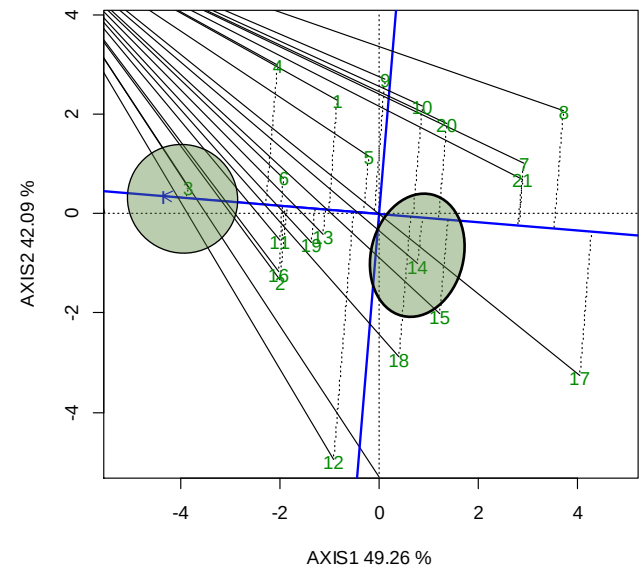
LR	DTH	DTM	Ph	SL	NSPMS	SPS	GPS	TSW	TGW	GY
BL 5106 (1)	71.33	95.00	55.31	5.38	162.33	8.67	15.33	6.08	26.08	523.07
BL 5099 (2)	72.67	95.00	45.91	6.08	156.33	10.00	16.00	6.85	29.31	803.47
BL 4984 (3)	72.00	95.33	51.00	6.48	138.00	7.67	14.67	6.10	26.92	655.83
Bhrikuti (4)	69.33	93.00	46.30	5.81	122.33	7.33	13.00	5.31	26.36	573.17
NL 1437 (5)	68.00	93.00	46.63	8.63	156.00	8.33	13.00	4.87	28.34	325.20
NL 1402 (6)	67.33	93.00	49.93	6.12	175.67	9.00	14.67	8.60	27.85	562.77
Gautam (7)	69.33	93.00	57.31	6.87	180.67	8.67	15.33	8.00	34.68	354.66
BL 5116 (8)	70.67	95.33	49.18	6.27	227.00	8.13	13.33	6.91	32.57	375.57
NL 1492 (9)	73.33	95.00	52.22	5.87	171.00	8.23	13.67	7.41	31.80	521.13
NL 1488 (10)	68.67	93.67	57.32	7.87	175.33	9.83	17.00	8.85	35.34	478.90
NL 1447 (11)	67.33	93.00	58.20	7.86	171.67	11.00	21.67	10.90	32.69	744.85
NL 1445 (12)	74.00	97.00	57.25	6.45	155.33	8.33	18.00	9.18	35.01	809.30
NL 1506 (13)	70.67	95.33	57.82	7.25	125.00	9.33	19.33	13.02	34.17	517.81
NL 1504 (14)	74.67	98.67	64.45	7.67	137.00	12.67	22.67	13.62	36.95	623.02
NL 1503 (15)	72.67	97.00	53.07	6.84	156.00	11.33	24.67	10.90	33.17	464.00
NL 1501 (16)	66.00	93.00	60.77	8.20	158.00	11.00	22.33	10.57	33.29	651.83
Vijaya (17)	64.00	91.00	59.51	6.81	144.67	8.67	16.67	7.58	32.26	707.07
NL 1512 (18)	67.33	93.00	60.52	7.33	159.67	10.00	21.00	9.93	30.78	856.63
NL 1509 (19)	66.00	91.67	55.41	6.56	163.00	10.67	20.67	9.34	31.35	652.00
NL 1508 (20)	74.67	96.00	49.53	6.10	176.33	10.00	20.00	7.67	33.60	600.00
Zinc Gahu 1 (21)	68.67	94.67	48.42	7.93	133.67	9.00	17.33	7.49	33.54	615.47
Mean	69.94	94.37	54.10	6.87	159.29	9.42	17.63	8.53	31.72	591.23
STD	3.08	1.89	5.38	0.89	23.00	1.36	3.55	2.35	3.13	146.87
CV	4.40	2.00	9.95	12.95	14.44	14.41	20.12	27.48	9.88	24.84
F-Value	***	***	ns	ns	ns	ns	ns	**	ns	ns



RESULTS AND DISCUSSION

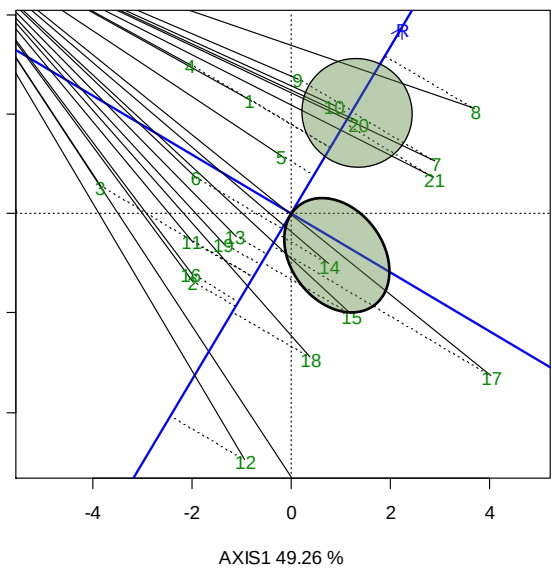


Examine an Environment



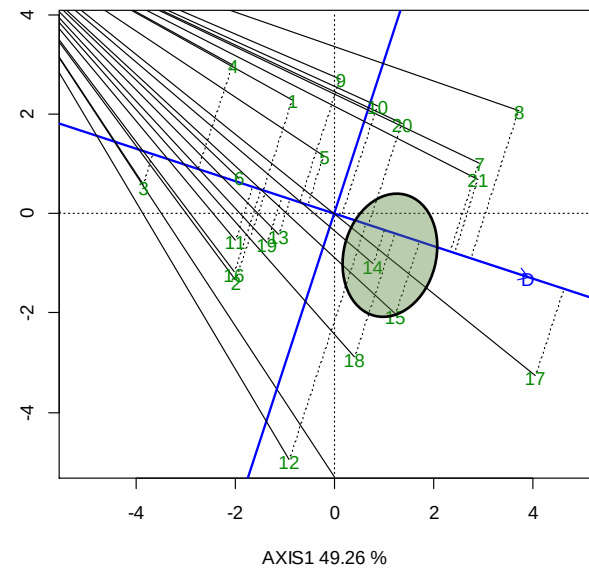
Irrigated

Examine an Environment



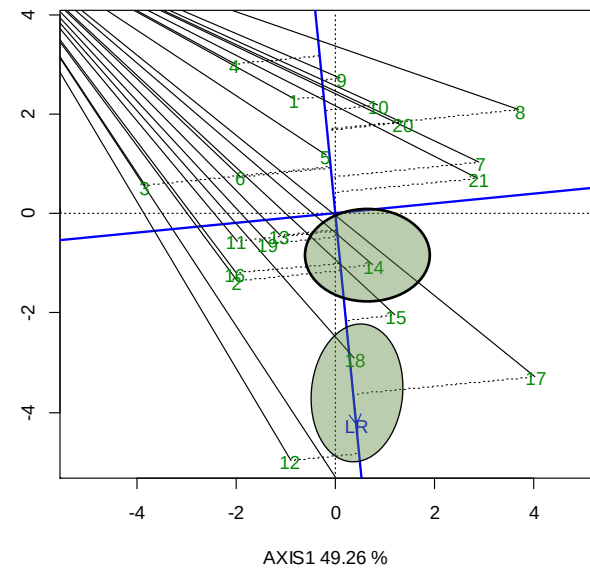
Rainfed

Examine an Environment



Drought

Examine an Environment

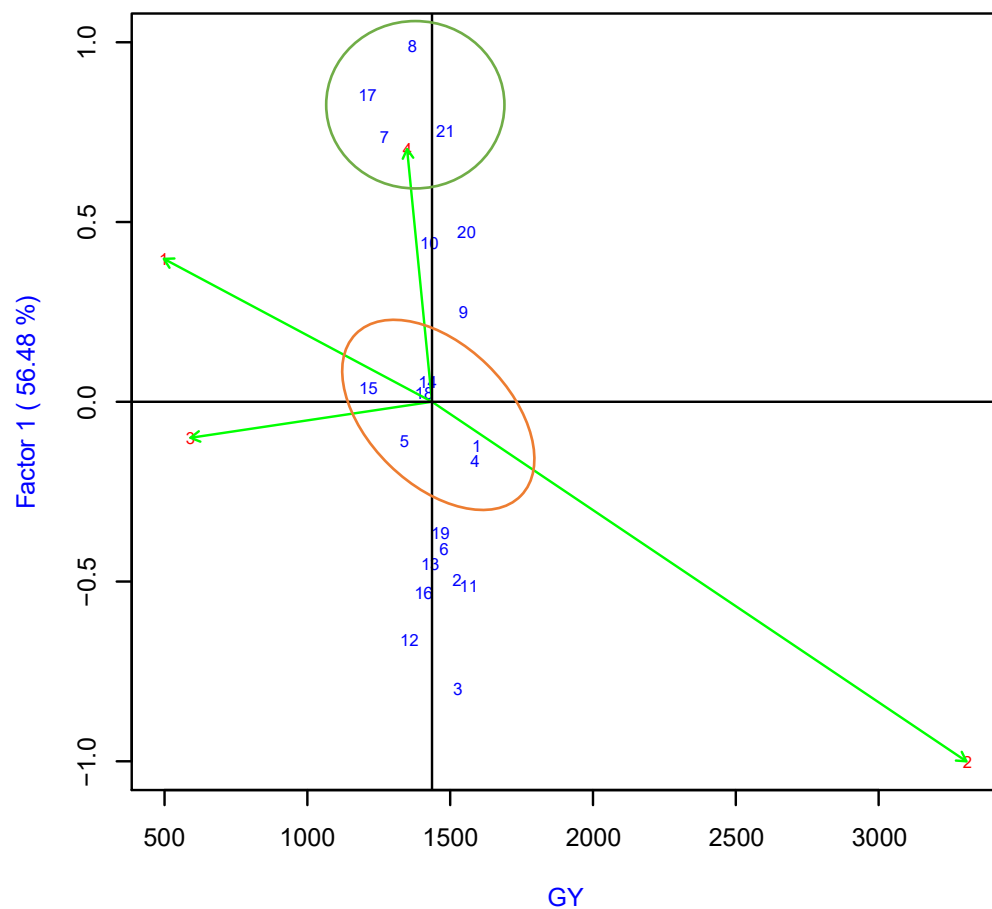


Late sown Rainfed



RESULTS AND DISCUSSION

AMMI PCA1 Score vs GY from a RCB



Stable genotype: 14, 18

	ENV	GEN	G X E
GY	96.58***	0.94	2.49

	PC1	PC2	Total
GY	56.48	37.95	94.43

The importance of first two IPCAs in stability and adaptability consistent with earlier studies in cereals (Mohammadi et al., 2018b; Neisse et al., 2018)

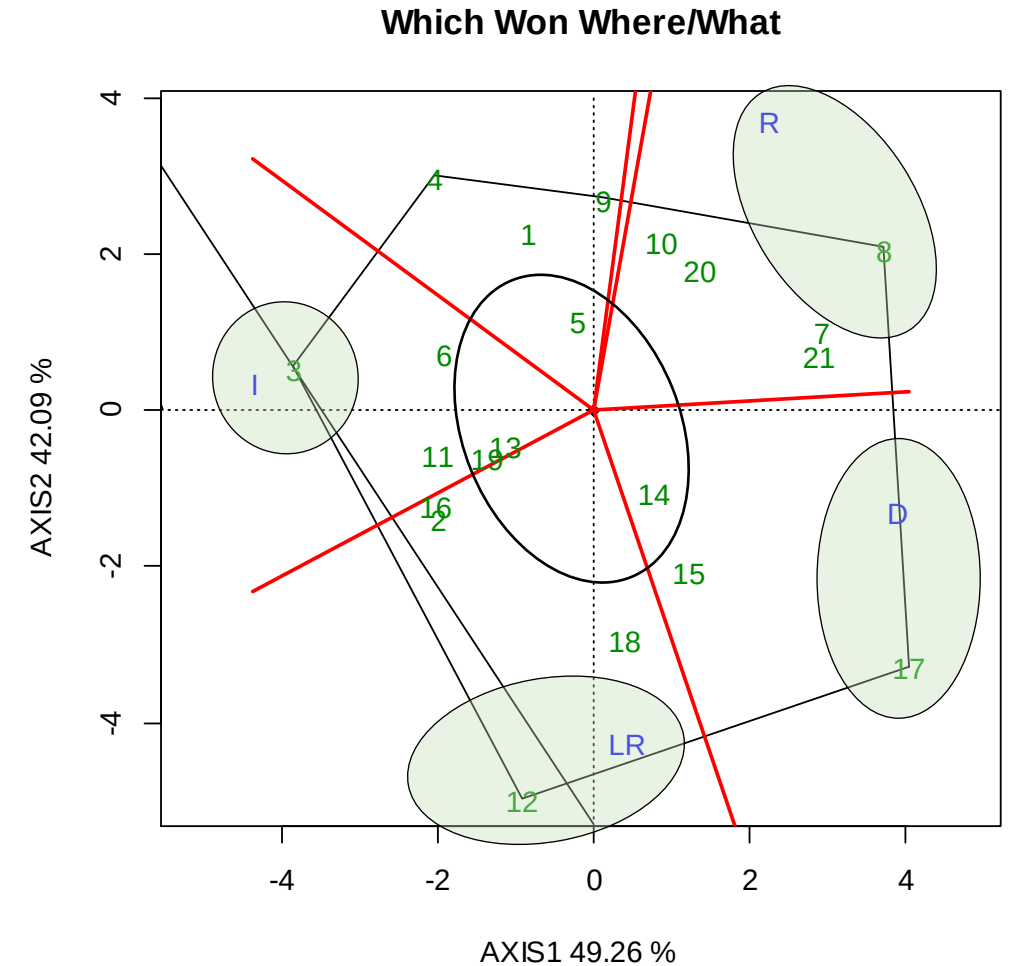
Dominance of environmental effects consistent with findings in South Asian wheat trials (Bhandari et al., 2024c; Bhandari and Poudel, 2024; Bishwas et al., 2021; Chaudhary et al., 2023; Gupta et al., 2022; Singh et al., 2019a)



RESULTS AND DISCUSSION



- Stable Genotype: 5, 14 (**WWW**), 14, 18 (**AMMI**)
- Complementarity among AMMI and GGE aligns with earlier studies
 - wheat (Thungo et al., 2019),
 - rice (Hasan et al., 2022; Senguttuvel et al., 2021), common vetch (Greveniotis et al., 2021)
 - chickpea (Erdemci, 2018)
 - maize (Li et al., 2023; Wolde et al., 2018)
 - sweet potato (Ngailo et al., 2019)
 - watermelon (Dia et al., 2016)
- **Stable genotype: 5, 14**
- **Adaptable genotype:**
 - Irrigated: 3
 - Rainfed: 8
 - Drought: 17
 - Late sown Rainfed: 12

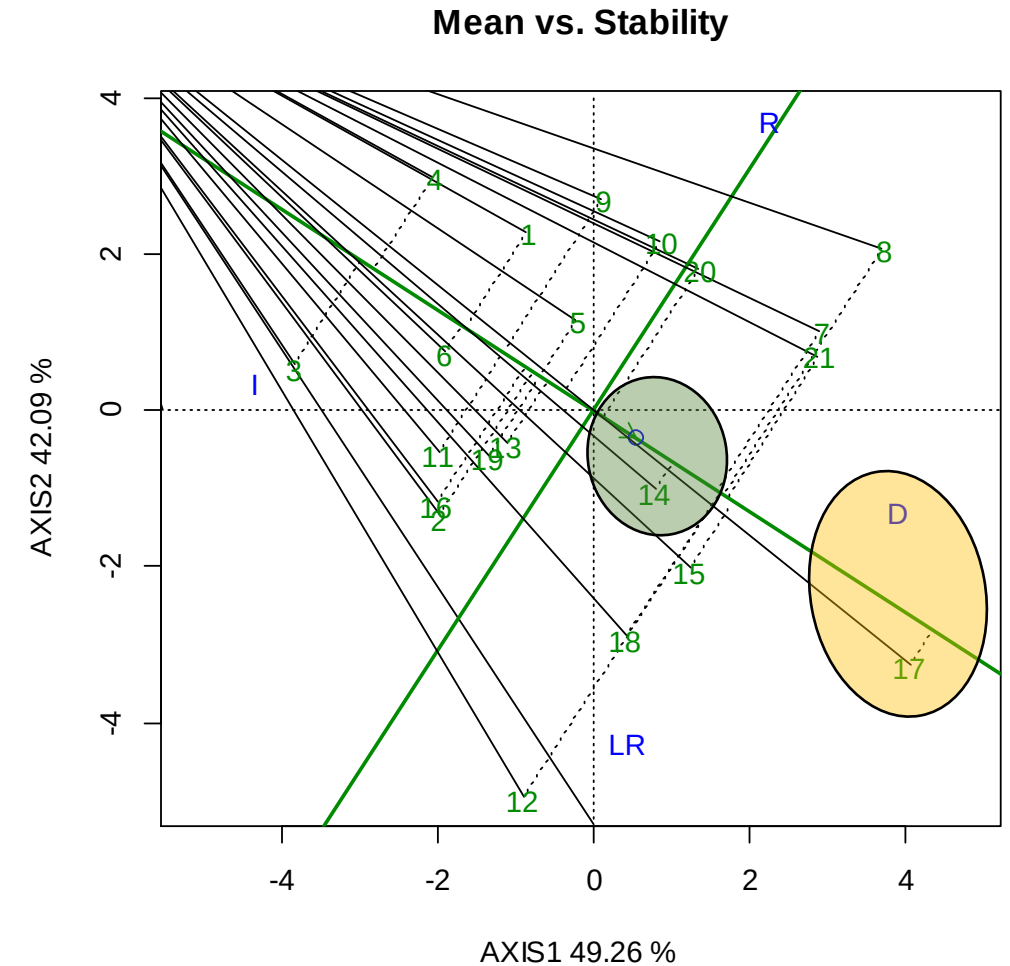




RESULTS AND DISCUSSION



- Mean vs. Stability model was also used by Apraku et al., 2016 and Aktas, 2016.
- High yielding: Lies towards the side of arrowhead (Ngailo et al, 2019).
- Stability: The length of projection on the AEC (Average Environment Coordinates) abscissa (Jat et al., 2017).
- **Above average yielding and stable: 14, 17**



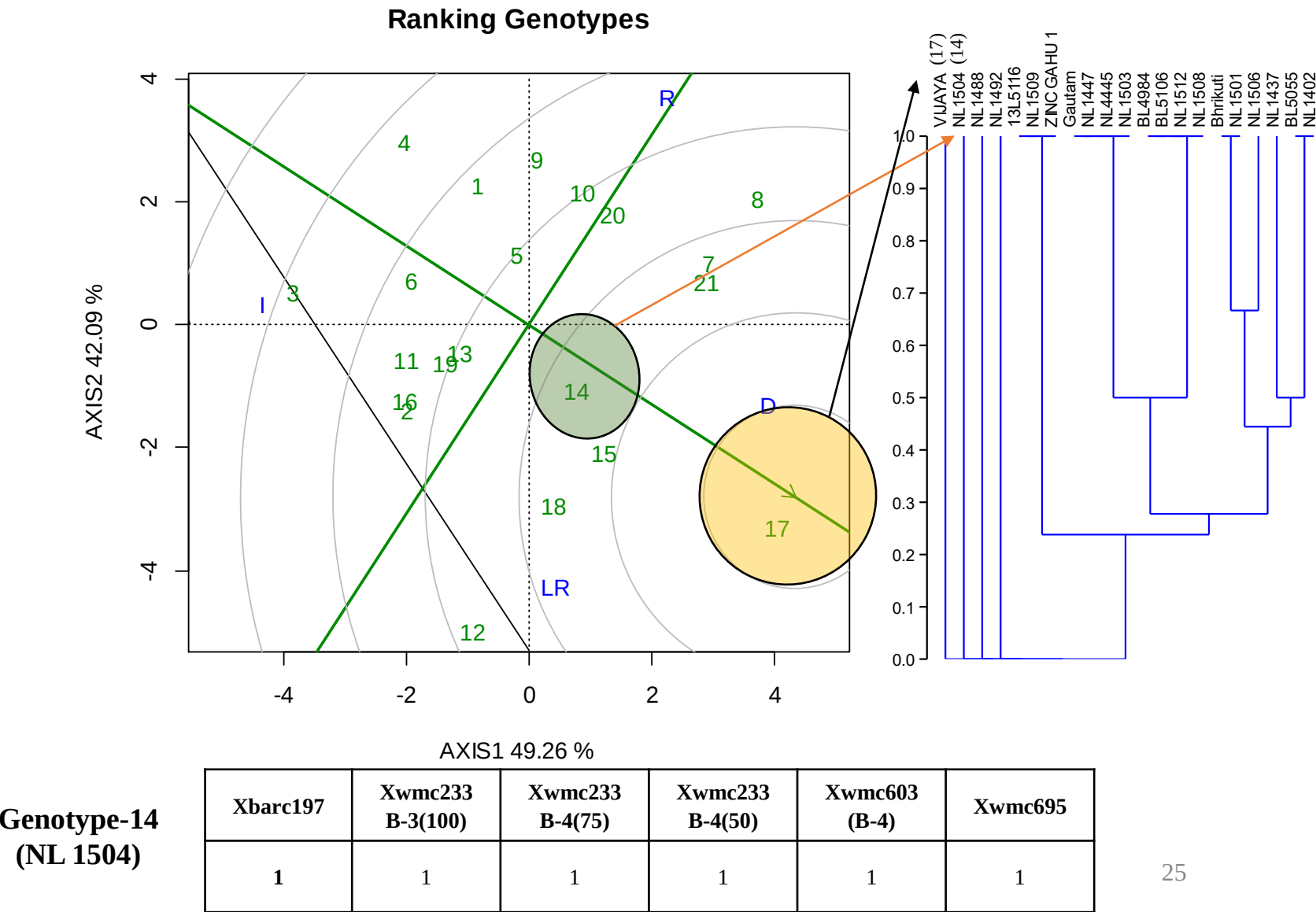
Such diversified trend of genotypes were also observed in previous studies (Bhandari et al., 2024c; Bishwas et al., 2021; Hashim et al., 2021; Khan et al., 2021a; Sabri et al., 2020; Y Oladosu, 2017)



RESULTS AND DISCUSSION



- Ideal Genotype: Least G x E effect
 - (Omrani et al., 2022)
 - (Ngailo et al., 2019)
 - (Yan and Kang, 2002)
- Ideal genotypes are located in the centre of the concentric circles (Bhandari et al., 2024a; Bishwas et al., 2021; Ngailo et al., 2019).
- **Ideal genotype: 17 (Vijay)**
- Ideal genotype selection using Ranking Genotype adapted
 - across rice (Hasan et al., 2022)
 - oats (Kebede et al., 2023b)
 - sugarcane (Mahadevaiah et al., 2021)
 - and cassava (Tumuhimbise et al., 2014)

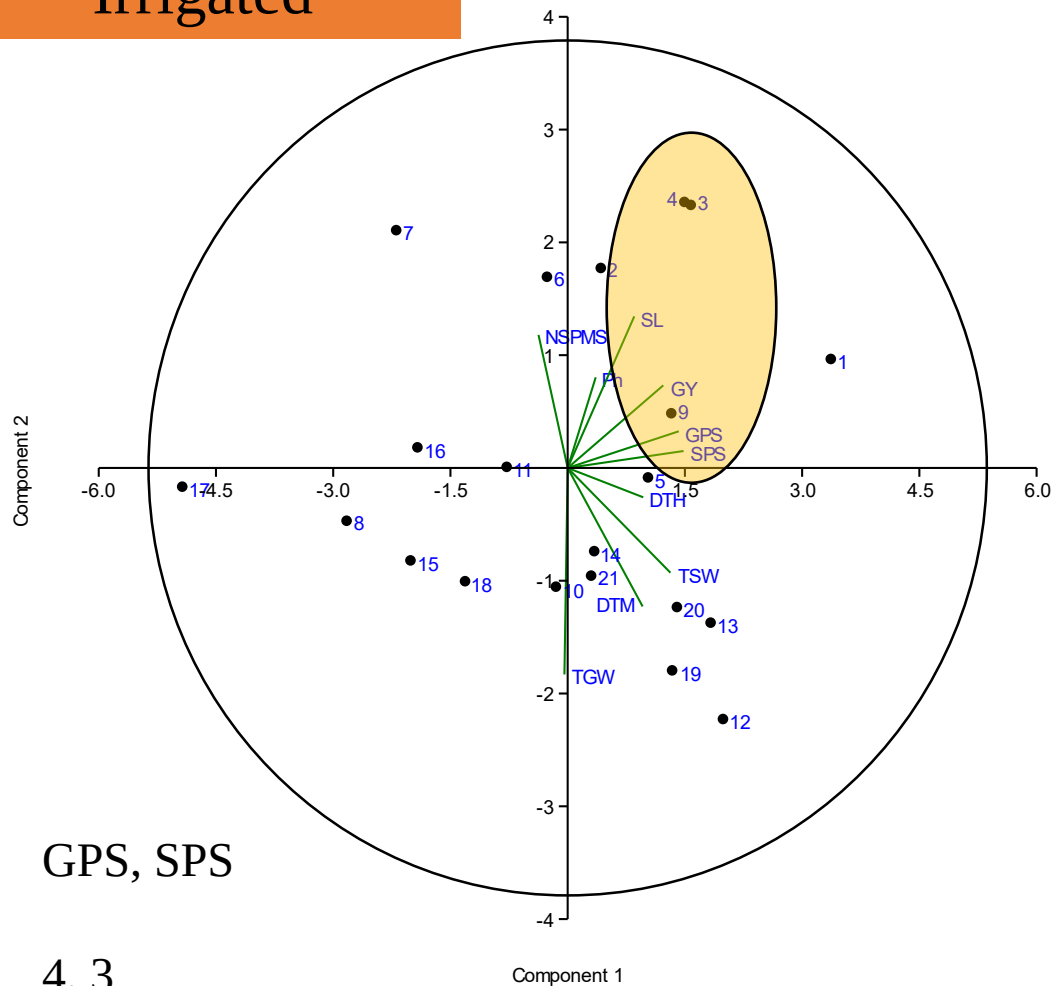




Trait Based Selection (TBS)

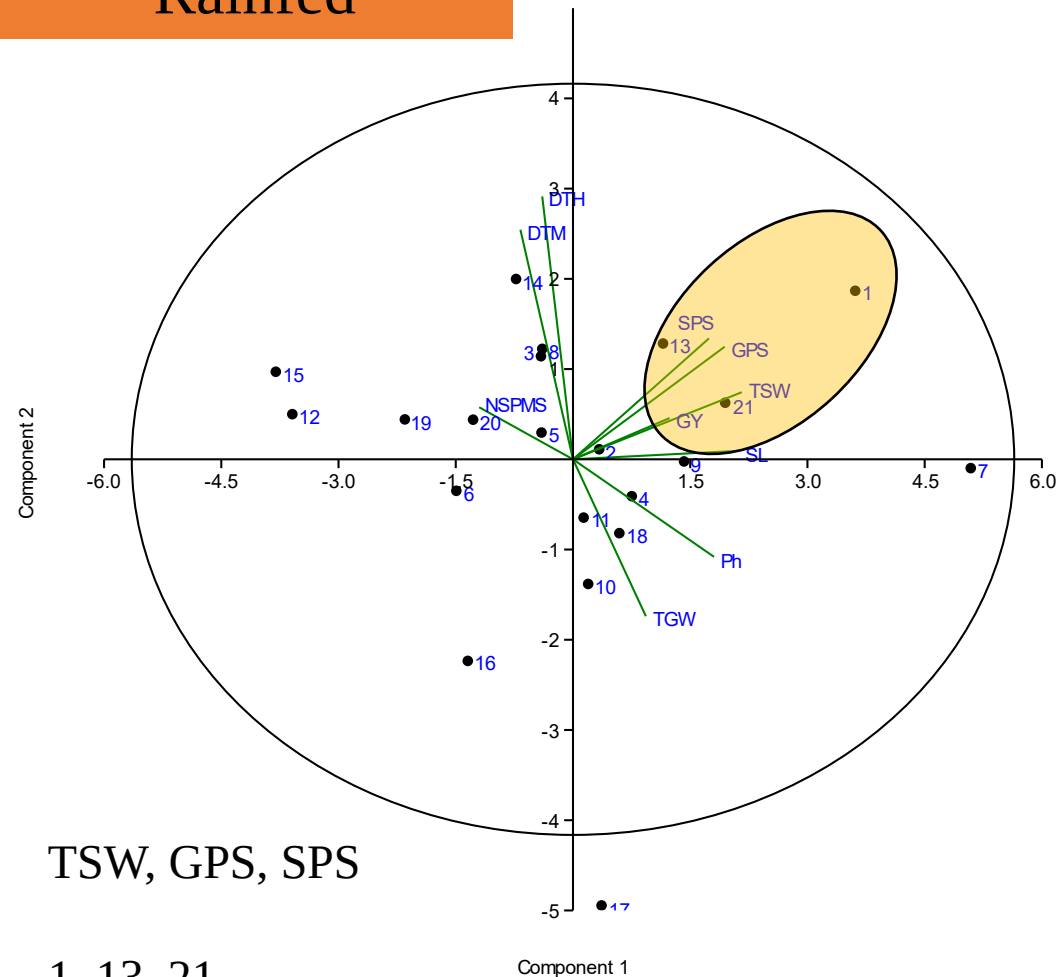


Irrigated



- GPS, SPS
- 4, 3
- Diverging: 17 (Vijay)

Rainfed



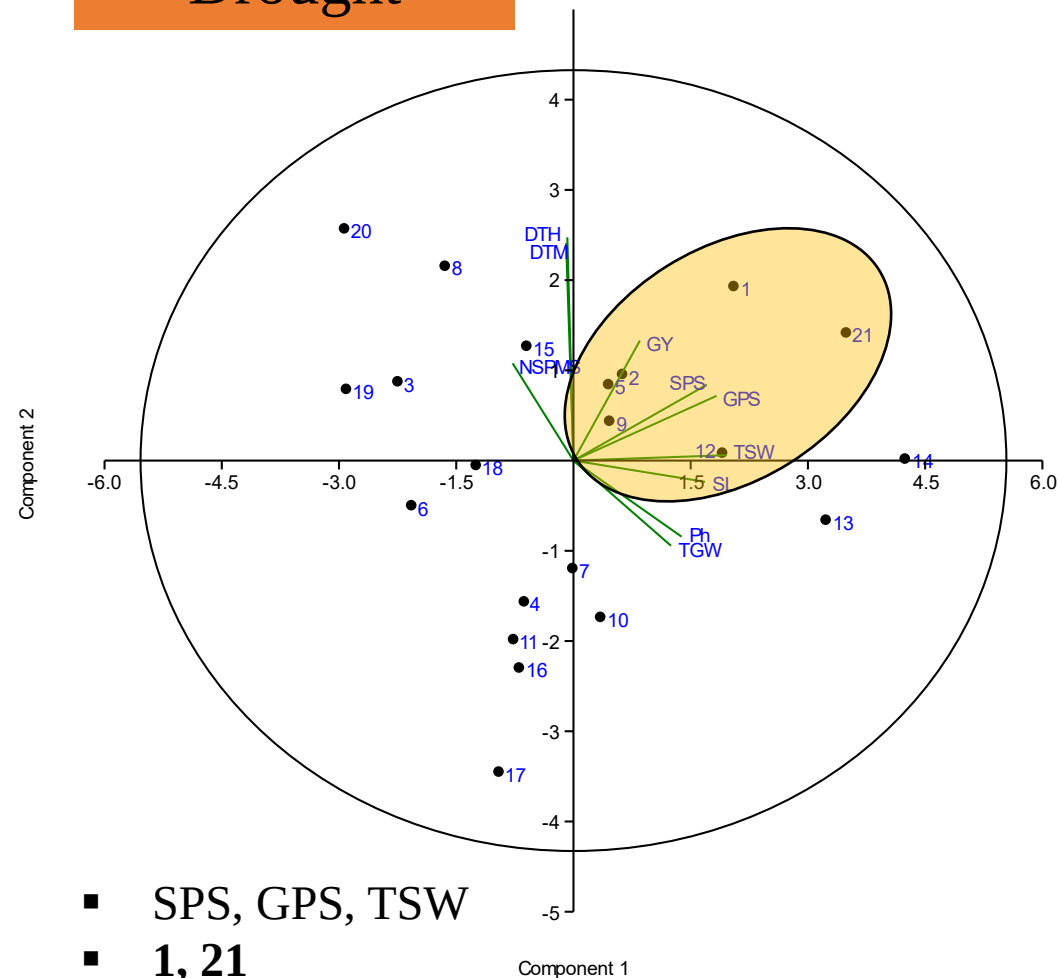
- TSW, GPS, SPS
- 1, 13, 21



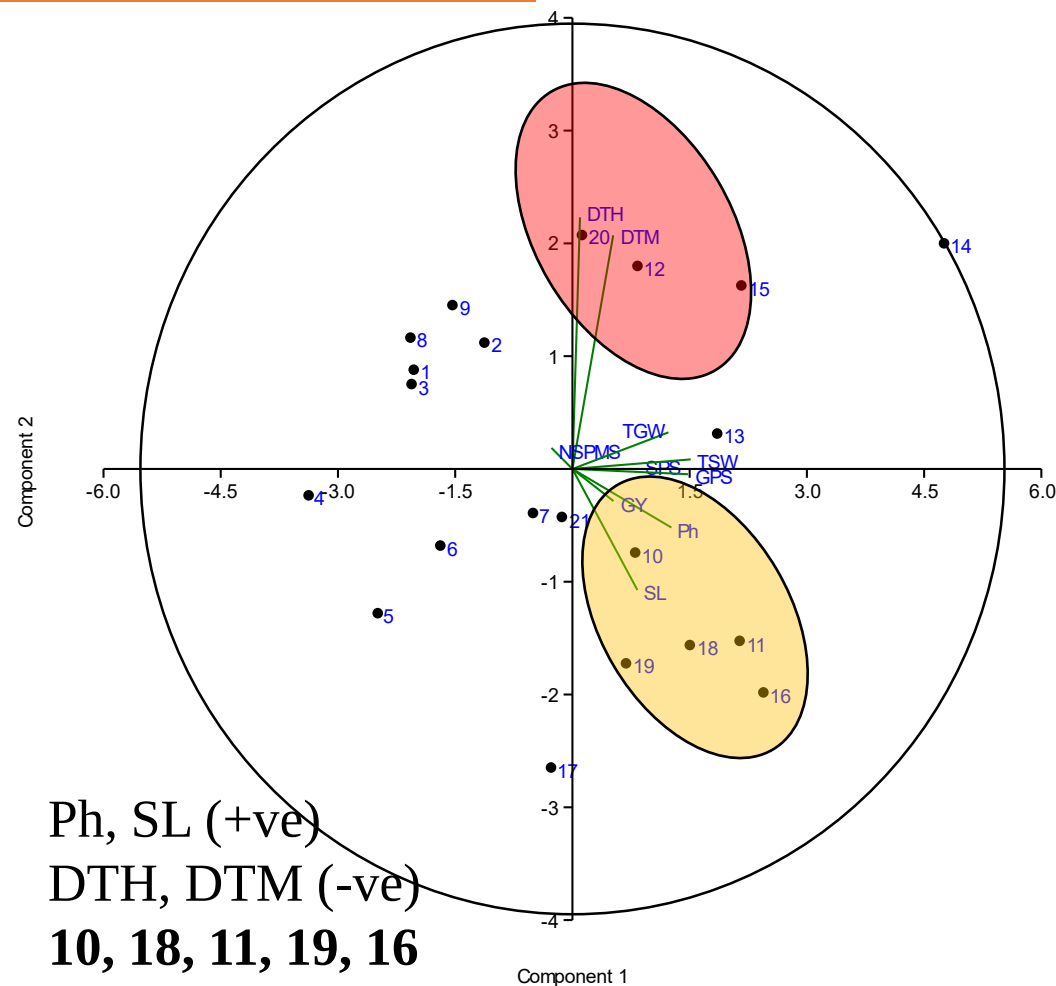
Trait Based Selection (TBS)



Drought



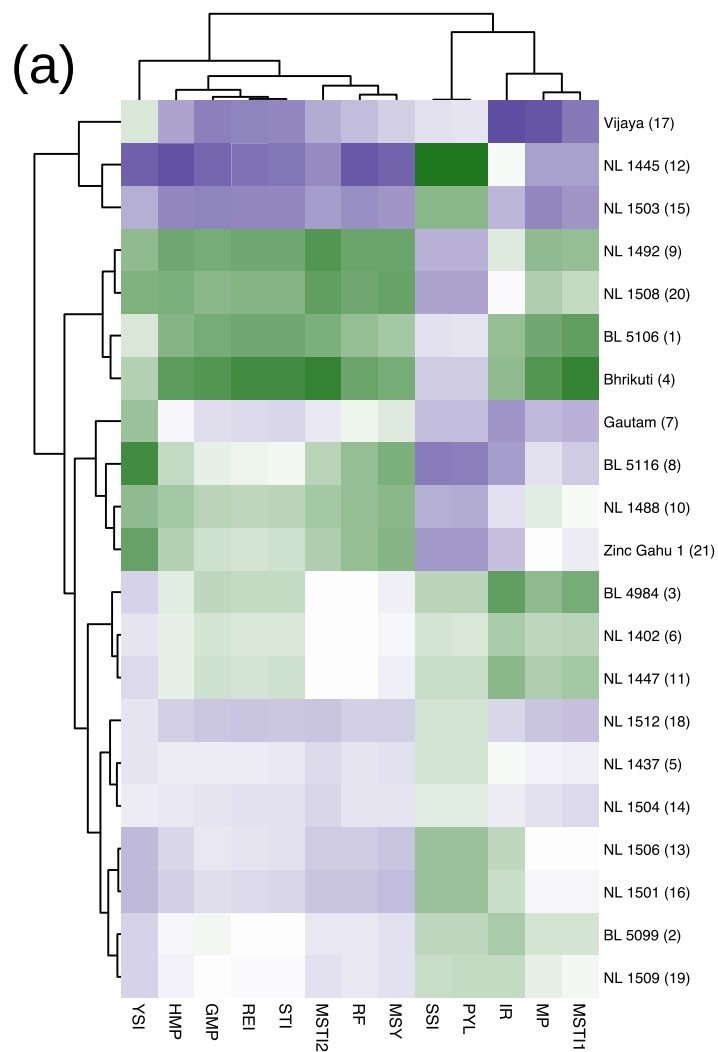
Late-Sown Rainfed



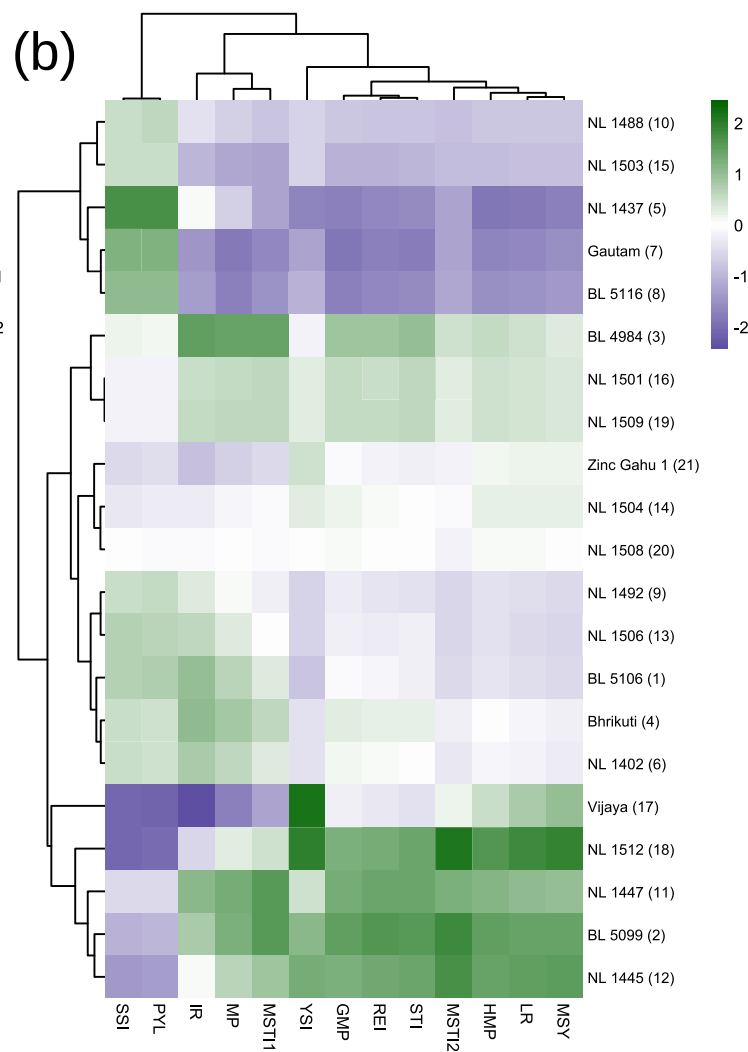
Validation of Earliness under late sown condition (Chen et al., 2016; Mondal et al., 2013; Pandey et al., 2015; Puri and Gautam, 2015; Yu et al., 2017)



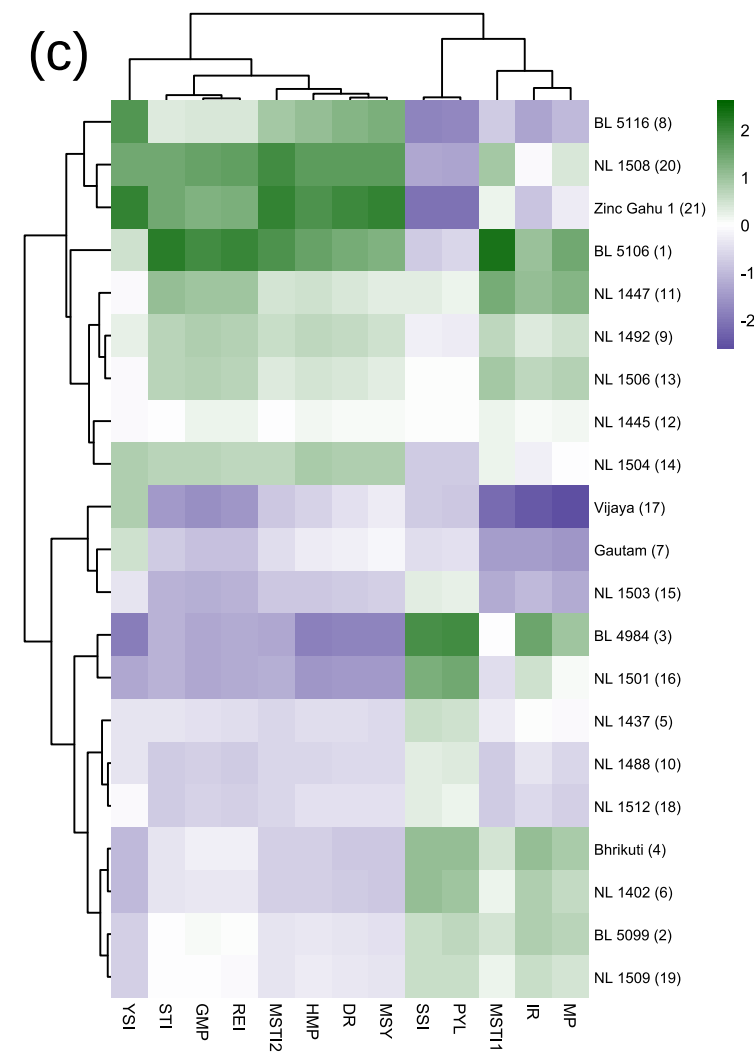
Validation of MSY and Varietal Performance



Irrigated-Rainfed (~60% Yield loss)



Irrigated-Late Rainfed (~80% Yield loss)



Irrigated-Drought (~85% Yield loss)

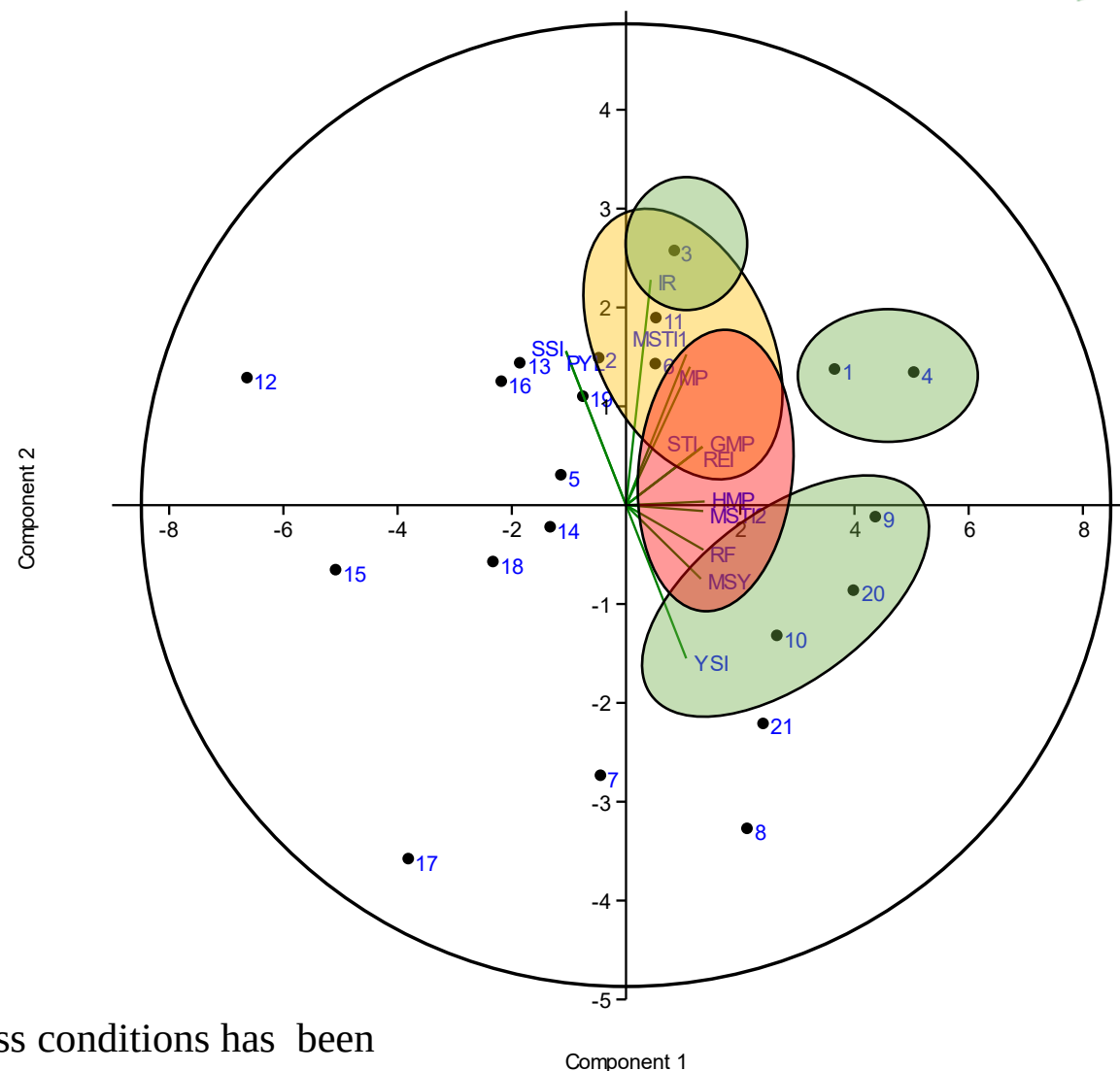


Validation of MSY



Irrigated-Rainfed (~60% PYL)

- Stress tolerance indices for Irrigated Yield (IR),
 - MST11
 - MP
 - Best Genotype: 3
- Stress tolerance indices for rainfed (RF) yield,
 - **Mean Stable Yield (MSY)**
 - Yield Stability Index (YSI)
 - Modified stress tolerance index 2 (MSTI2)
 - Harmonic mean productivity (HMP)
 - Best Stress tolerant Genotype: 2, 10, 8

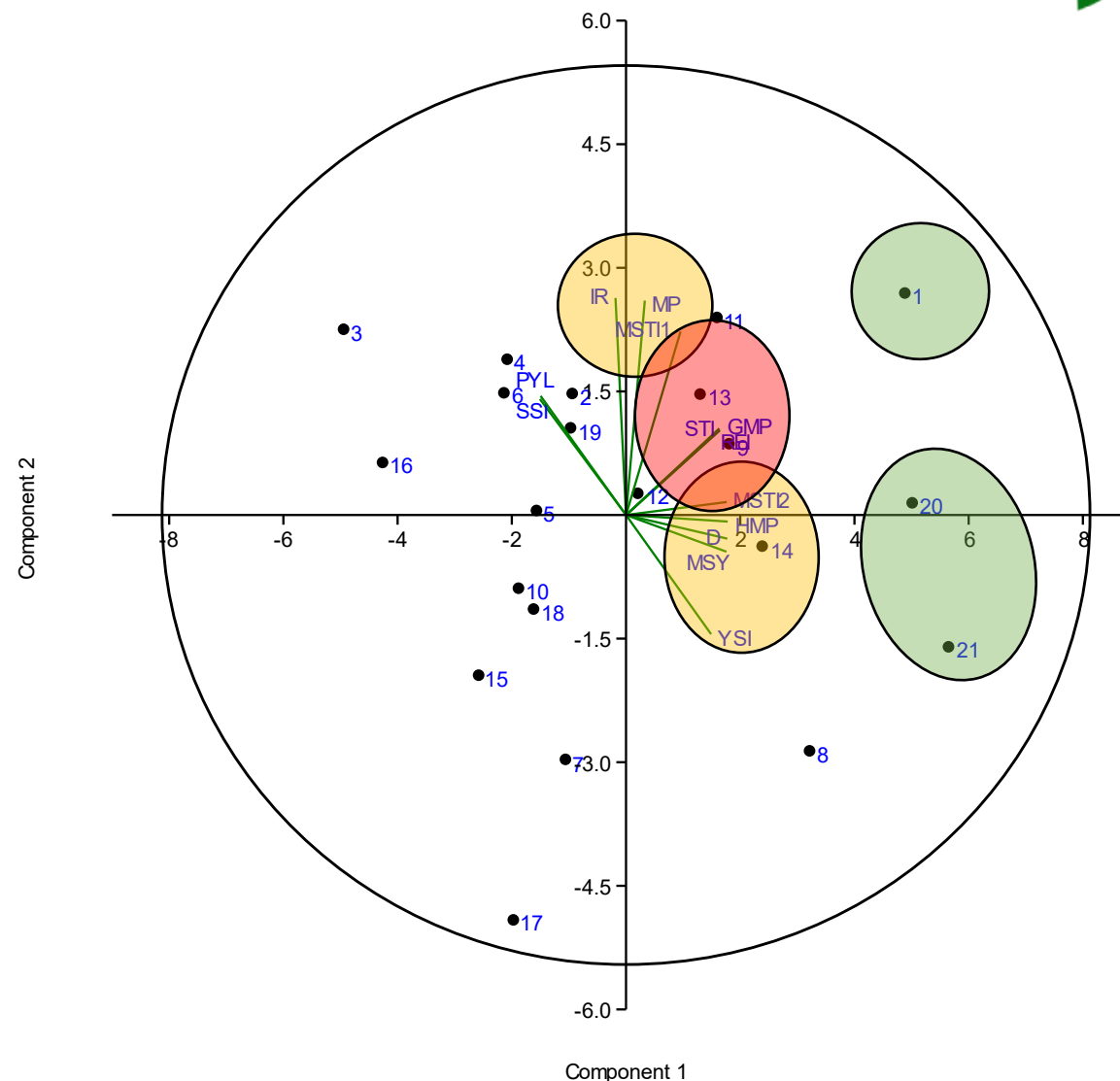


The clustering of GMP, STI and REI with both stress and non-stress conditions has been reported in barley (Taherian et al., 2017), maize (Khodarahmpour et al., 2011), wheat (Bhandari et al., 2024b; Kamrani et al., 2017b).



Irrigated-Drought (~85% PYL)

- Stress tolerance indices for Irrigated Yield (IR),
 - MSTI1
 - MP
- Stress tolerance indices for Drought (D) yield,
 - Mean Stable Yield (MSY)
 - Modified stress tolerance index 2 (MSTI2)
 - Harmonic mean productivity (HMP)
 - Yield Stability Index (YSI)
 - **Best Stress tolerant Genotype: 2, 14, 21**



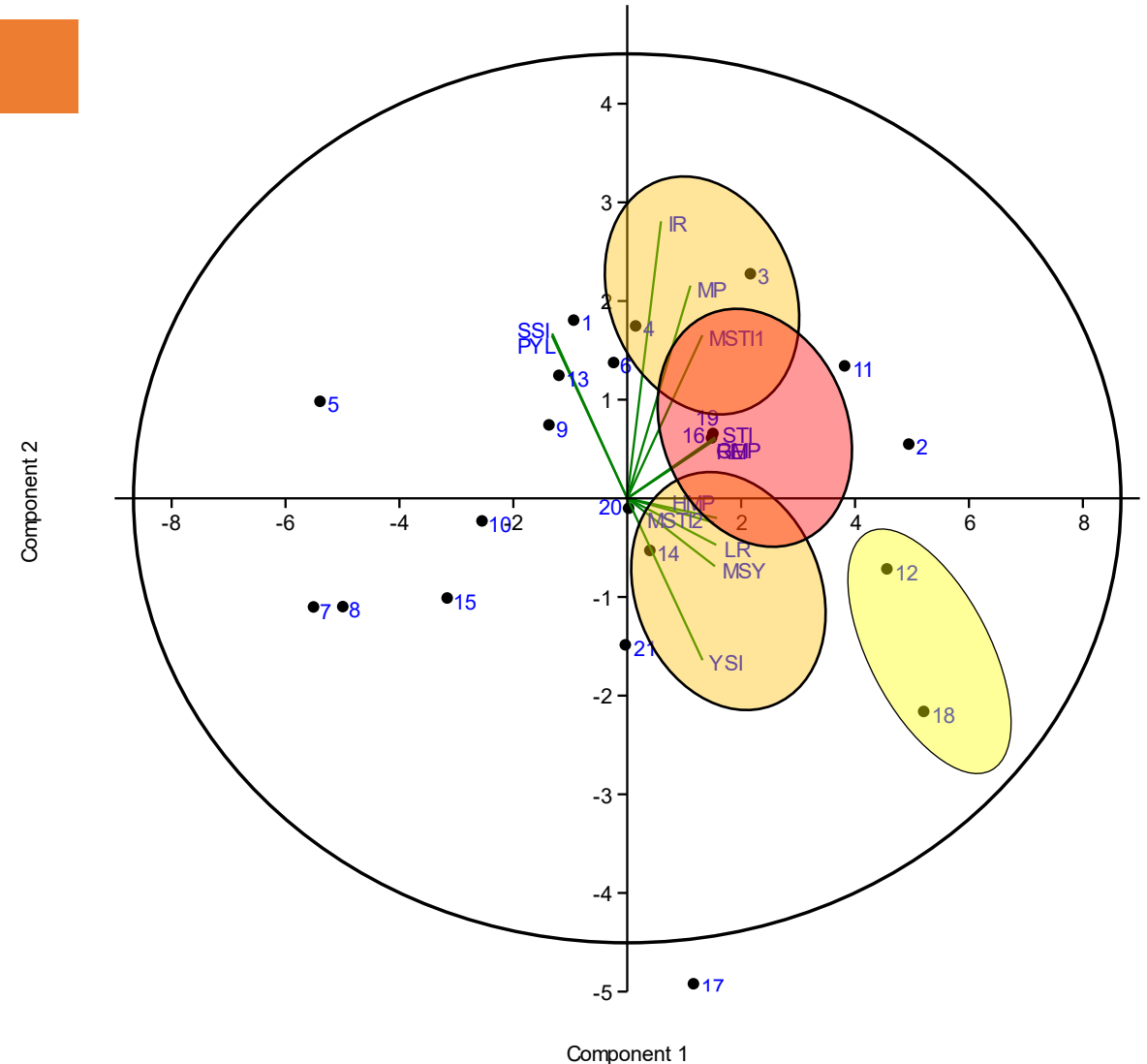


Validation of MSY



Irrigated-Late Sown Rainfed (~80% PYL)

- Stress tolerance indices for Late sown rainfed (LR),
 - MSTI1
 - MP
- Stress tolerance indices for Late Rainfed (LR) yield,
 - **Mean Stable Yield (MSY)**
 - Modified stress tolerance index 2 (MSTI2)
 - Harmonic mean productivity (HMP)
 - Yield Stability Index (YSI)
 - **Best Stress tolerant Genotype: 12,18**





MOLECULAR VALIDATION



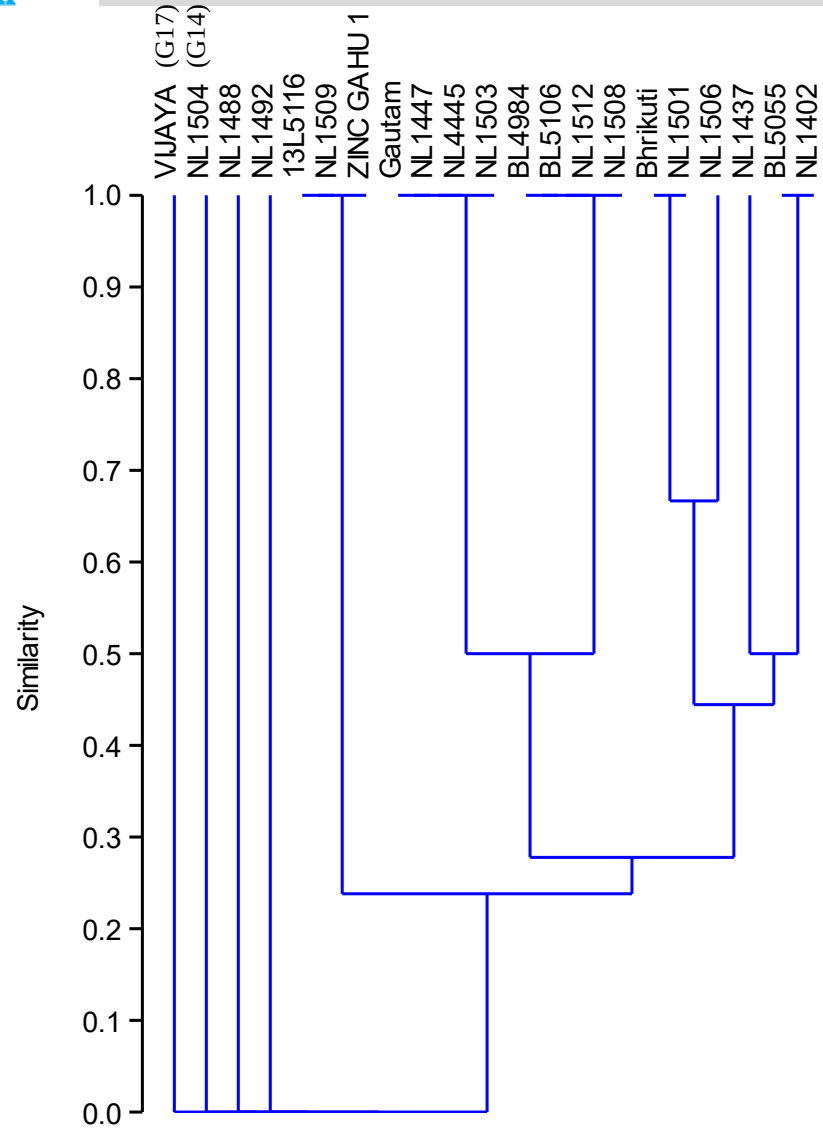
Marker	Primer sequence	Tm (C)	Scored Bands	Polymorphis m (%)	PIC Value
wms46	F- GCA CGT GAA TGG ATT GGA C	52	1 (100bp)	100%	0.50
	R- TGA CCC AAT AGT GGT GGT CA				
wms130	F- AGC TCT GCT TCA CGA GGA AG	56	1 (100bp)	100%	0.50
	R- CTC CTC TTT ATA TCG CGT CCC				
wms262	F- AAA CCC TTG CAC AAA TC GA	52	1 (B-3)	100%	0.41
	R- TTT AGC TAC AGG AGG CC				
Xbarc197	F: CGCATGGTCAGTTTTCTTTTAATCC	54	1 (75 bp)	100%	0.25
	R: GCGCTCTCCTTCATTTATGGTTTGTTG				
Xwmc233	F-GAC GTC AAG AAT CTT CGT CGG A	52	3 (100, 75, 50 bp)	100%	0.43*
	R-ATC TGC TGA GCA GAT CGT GGTT				
Xwmc603	F-ACA AAC GGT GAC AAT GCA AGG A	54	5 (B-1 to B-5)	20%	0.31**
	R-CGC CTC TCT CGT AAG CCT CAA C				
Xwmc695	F-GAG GGC ACC TCG TAA GTT GG	54	1 (75 bp)	100%	0.25
	R-GGC AGG AGC CCC TAC AAG AT				

*Average PIC value for the three polymorphic loci (0.31, 0.49, and 0.49).

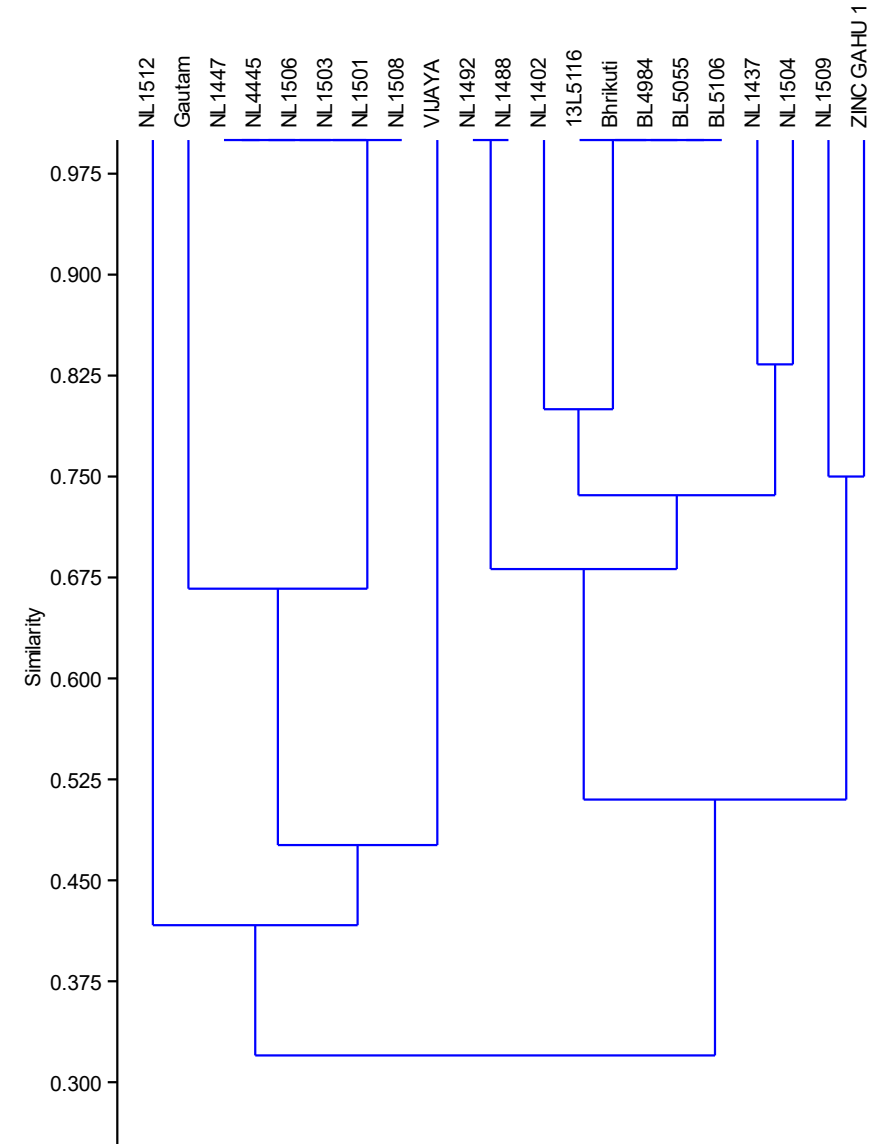
**PIC value for the single polymorphic band (B-4); monomorphic bands (B1-B3, B5) have a PIC of 0.00.



MOLECULAR VALIDATION



UPGMA based Clustering by Using
Drought based SSR markers



UPGMA based Clustering by Using
Heat based SSR markers



CONCLUSION

Severe Yield Reduction
rainfed)
Environmental sensitivity of wheat

59.20% (RF) 84.95% (D) 82.12 (Late sown

(96.58***)% variance explained by ENV component

Obj-2 Trait Based Selection:

GPS, SPS, TSW (All)
Plant height and earliness (Late sown heat)

Obj-1 Highest Performing Genotypes

Stable: 5, 14 (Genotype
with validated gene)
Adaptable (Irrigated) 3
Adaptable (Rainfed) 8
Adaptable (Drought) 17 (Vijay)
Adaptable (Late sown rainfed) 12

Obj-3 Validation of Selection Indices:
Predictive Power:
Recommended Indices

MSTI1, MP (Irrigated)
MSY, MSTI2, HMP (RF, D, LR)

Obj-4 Genotype-14 (NL 1504)

Xbarc19
7

Xwmc23
3
B-3(100)

Xwmc23
3
B-4(75)

Xwmc23
3
B-4(50)

Xwmc60
3
(B-4)

Xwmc69
5



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Thank you !